RL - Lab - 1

Q1:- Arithmetic mean

A:- $Q(A) = \frac{3+2+4}{3} = \frac{9}{3} = 3$

B:- $Q(B) = \frac{6+7}{2} = \frac{13}{2} = 6.5$

C:- $Q(C) = \frac{5+4+6}{3} = \frac{15}{3} = 5$

$6.5 > 5 > 3$

∴ Greedy Selected machine = B

Q2:- Incremental update

$$Q_{new} = Q_{old} + \frac{R - Q_{old}}{N}$$

A:- $0 + \frac{3-0}{1} = 3$

$3 + \frac{2-3}{2} = 2.5$

$2.5 + \frac{4-2.5}{3} = 3$

B:- $0 + \frac{6-0}{1} = 6$

$6 + \frac{7-6}{2} = 6.5$

C:- $0 + \frac{5-0}{1} = 5$

$5 + \frac{4-5}{2} = 4.5$

$4.5 + \frac{6-4.5}{3} = 5$

Finally $Q(A) = 3$, $Q(B) = 6.5$, $Q(C) = 5$

Greedy Selection :- B